

SCENARIO

A firecracker thrown into a busy plaza causes a brief panic. Pedestrians initially crowd around the disturbance, then disperse radially in nearly straight paths toward the open exits before the flow gradually returns to a calm walking pace.

TrajFacts
HUMAN TRAJECTORY
KNOWLEDGE



LLM
BEHAVIORAL
DECOMPOSITION

V R D S T
PROTOCOL

QUESTIONS

Q1. Are people running in panic?

mean_speed

Q2. Are people running away from the firecracker?

directional_entropy

flow_alignment

path_linearity

Q3. Do people first run fast then slow down after they run away?

speed_trend

density_trend

ANSWERS

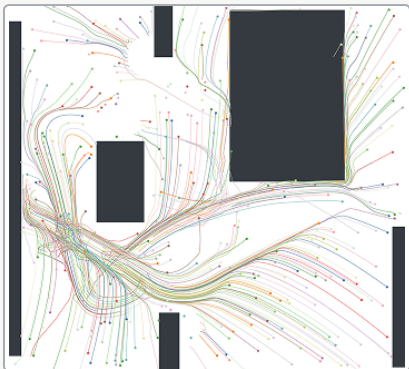
0.55 0.92

0.42 0.80

0.48 0.82

Stage 1 · STRIDE-Bench Generation

TRAJECTORY



Q1. Are people running in panic?

Q-SCORE 1.00

mean_speed

1

Q2. Are people running away from the firecracker?

Q-SCORE 0.67

directional_entropy

1

flow_alignment

1

path_linearity

0

Q3. Do people first run fast then slow down after they run away?

Q-SCORE 1.00

speed_trend

1

density_trend

1

SCENE SCORE

STRIDE_{scene} = 0.89

$= (1.00 + 0.67 + 1.00) / 3$

$Q_i = \text{mean}_m(s_{im})$
 $\text{STRIDE} = \text{mean}_i(Q_i)$

Stage 2 · STRIDE Score Calculation

DMT Set

Domain Measurement
Toolkit, organised by
VRDST

V VELOCITY

mean_speed

speed_variation_coeff

R REALISM

collision_fraction

lingering_fraction

D DIRECTION

flow_alignment

path_linearity

S SPATIAL

peak_local_density

dispersal_score

T TEMPORAL

speed_trend

density_trend